Tech Knowledge Share

Bagging & Boosting

Demystifying Ensemble Learning Strategies for High-Performance Machine Learning

By **Venu Gopala Krishna Vanka**

Ensemble Learning Foundations

Combining weak base learners to construct a highly accurate, robust, and generalizable predictive model.

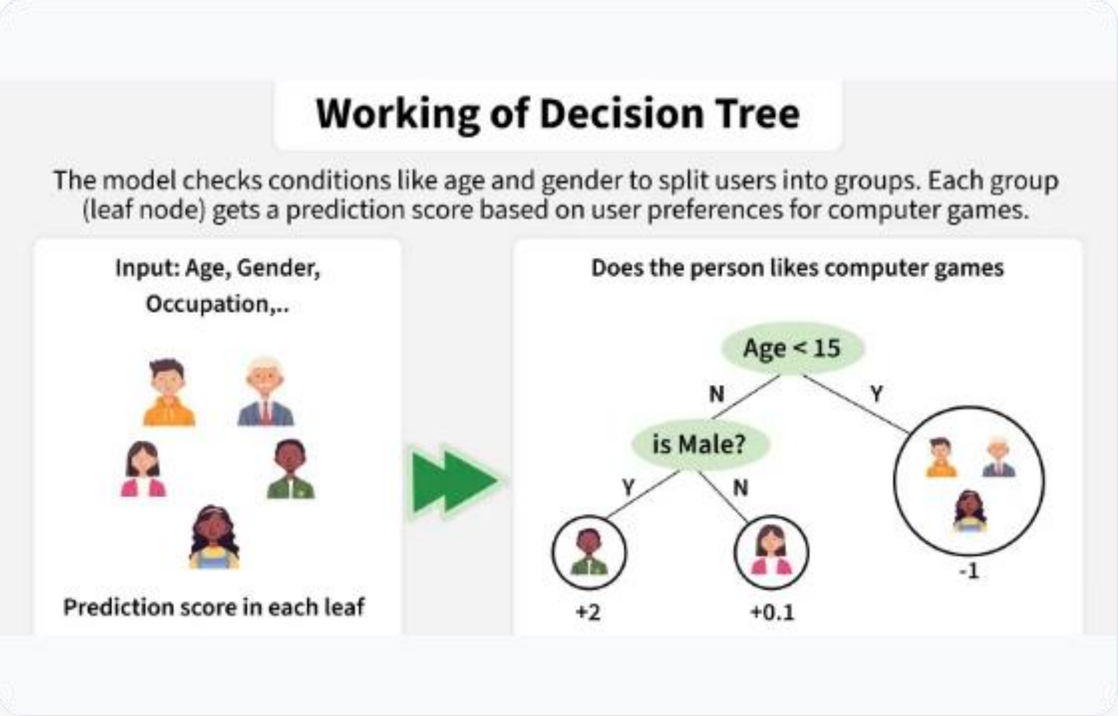
Core Concept: Ensemble Models

What is Ensemble Learning?

The primary purpose of using an ensemble model is to group a set of weak learners and aggregate their predictions to form a single strong learner.

> Total Error = Bias + Variance + Noise

- **Weak Learner:** A model that performs slightly better than random guessing.
- **Strong Learner:** A combined model achieving arbitrary high accuracy.
- **Optimization Goal:** Control both bias and variance to minimize generalization error.



Bagging vs. Boosting At a Glance

Parameter	Bagging	Boosting
Primary Objective	Decreases Variance (solves Overfitting)	Decreases Bias (solves Underfitting)
Data Partitioning	Random Bootstrap Sampling with Replacement	Higher weight given to misclassified samples
Goal to Achieve	Minimum Variance & Stable Predictions	Maximum Accuracy via Sequential Corrections
Sub-space Strategy	Random Subspace Sampling	Gradient Descent / Sequential Loss Minimization
Aggregation Method	Simple / Weighted Averaging	Weighted Majority Vote
Classic Example	Random Forest Classifier	AdaBoost, XGBoost, Gradient Boosting

Understanding Bagging

Objective & Mechanism

Bagging (Bootstrap Aggregating) is designed to reduce the variance of a decision tree classifier without altering bias.

- Creates several random subsets of data from the training set chosen **with replacement**.
- Each individual subset is used to independently train its own decision tree model.

Robust Predictions

Instead of relying on a single complex decision tree, predictions from all trained trees are aggregated.

- **Classification:** Uses majority voting across all generated trees.
- **Regression:** Uses the average of all predictions.
- Significantly reduces overfitting and model variance.

The 4-Step Bagging Process

1. Bootstrap Sample

Sample training dataset randomly with replacement given N observations and M features.

2. Feature Selection

Randomly select a subset of m features; use the best feature split iteratively.

3. Tree Growth

Grow each individual decision tree fully to its largest depth without pruning.

4. Aggregation

Repeat steps n times; aggregate predictions via majority voting or mean average.

Bagging: Pros & Cons

Advantages

- **Overfitting Reduction:** Effectively reduces model variance and prevents overfitting.
- **High Dimensionality:** Handles high-dimensional feature spaces exceptionally well.
- **Missing Data:** Maintains accuracy even when datasets contain missing observations.

Disadvantages

- **Mean Dependency:** Because final predictions rely on subset tree averages, it may lack precise local boundary values.
- **Interpretability:** Loss of simple decision tree interpretability (Black-box ensemble nature).

Python Implementation: Bagging

Random Forest Syntax

Random Forest implements bagging by constructing multiple decision trees using bootstrap aggregation and random feature subsets.

80 Trees

Optimal Ensemble Estimators

```
# Import Random Forest Classifier from sklearn.ensemble import RandomForestClassifier # Initialize Ensemble Model rfm = RandomForestClassifier ( n_estimators =80, oob_score =True , n_jobs =-1, random_state =101, max_features =0.50, min_samples_leaf =5 ) # Fit and Predict rfm.fit (x_train, y_train) predicted = rfm.predict_proba (x_test)
```

Understanding Boosting

Sequential Learning

Boosting creates a collection of predictors sequentially. Early weak learners fit simple models, and subsequent models actively analyze and fix prior errors.

When samples are misclassified, their weights are increased so following hypotheses prioritize classifying them correctly.

The 4-Step Boosting Process

1. Initial Subset (d1)

Draw random sample subset d1 without replacement to train first weak learner C1.

2. Weighted Focus (d2)

Draw subset d2 & add 50% misclassified samples from C1 to train weak learner C2.

3. Disagreement (d3)

Identify training samples where C1 and C2 disagree to train weak learner C3.

4. Majority Vote

Combine all sequential weak learners via weighted majority voting scheme.

Boosting: Pros & Cons

Advantages

- **Flexible Loss Functions:** Supports versatile loss functions (e.g., 'binary:logistic').
- **Complex Interactions:** Captures high-order feature interactions effortlessly.
- **Higher Accuracy:** Often achieves superior predictive accuracy over single models.

Disadvantages

- **Overfitting Risk:** Prone to overfitting if noise or outliers are heavily weighted.
- **Hyperparameter Sensitivity:** Requires careful tuning of learning rate, depth, and estimators.

Image Sources

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